Late Laramide dolomite recrystallization of the Husky Rainbow "A" hydrocarbon Devonian reservoir, northwestern Alberta, Canada: paleomagnetic and geochemical evidence

Michael T. Lewchuk, Ihsan S. Al-Aasm, David T. A. Symons, and Kevin P. Gillen

Abstract: The Rainbow Field is in reefal carbonates of the Middle Devonian Keg River Formation in the Western Canada Sedimentary Basin. Alternating field and thermal step demagnetization was done on specimens from unoriented core from a vertical, an inclined, and a horizontal well core in the dolomite reservoir. Although they had no viscous remanent magnetization component to use for orientation, most specimens had a well-defined characteristic remanent magnetization (ChRM) that resides in single to pseudosingle domain magnetite. By rotating the mean ChRM direction around its core axis, a small circle can be generated for each core and the small circles intersect in the true ChRM direction of D = 168°, I = –73.5° (α_{95} = 5.8°, k = 32.2). Its paleopole of 164°E, 83°N (A_{95} = 10°) defines a Tertiary age with one sigma limits of Middle Eocene to Middle Miocene age. Petrographic examination defines four generations of dolomite. Matrix dolomite has 60–100 μm diameter crystals that were later recrystallized to 200–400 μm. Dolomite cements are represented by vug-filling coarse dolomite (100–200 μm) and saddle dolomite (1000 μm). All four generations of dolomite give similar $\delta^{18}O$ values of –10.7 to –16.5‰ (Peedee Belemnite, PDB), $\delta^{13}C$ values of +0.7 to +3.2‰ (PDB), and Sr isotopic ratios of 0.70826 to 0.70846 that do not match the expected Middle Devonian carbonate or seawater values. We interpret these data to indicate that mixed pre-Laramide basinal fluids, heated by burial during the Laramide Orogeny, were present during late Laramide time when the dolomites were recrystallized and (or) precipitated prior to petroleum migration and accumulation in the Rainbow "A" reservoir. Thus the combined use of paleomagnetism, geochemistry, and petrography has been proven to be a useful technique to date and identify dolomitization events and pathways for the migration of hydrocarbons.

Résumé : Le Rainbow Field est situé dans des carbonates récifals de la Formation Keg River du Dévonien moyen dans le bassin sédimentaire de l'Ouest canadien. Une démagnétisation par champ alternatif et par étapes thermiques a été effectuée sur des spécimens de carottes non orientées prélevées d'un sondage vertical, d'un sondage incliné et d'un sondage horizontal dans le réservoir de dolomie. Quoique les carottes n'avaient pas de composante d'aimantation visqueuse rémanente pour servir d'orientation, la plupart des spécimens avaient une aimantation caractéristique rémanente bien définie (ChRM) qui réside dans une magnétite à moment unique ou pseudo-unique. En effectuant une rotation de la direction moyenne ChRM autour de l'axe de la carotte, un petit cercle peut être généré pour chaque carotte et ces petits cercles s'interceptent à la direction véritable du ChRM, D = 168°, I = –73,5° (α_{95} = 5,8°; k = 32,2). Son paléopole de 164°E, 83°N (A_{95} = 10°) définit un âge Tertiaire avec des limites de un sigma de l'Éocène moyen au Miocène moyen. Un examen pétrographique permet de déterminer quatre générations de dolomie. La dolomie de la matrice a des cristaux de 60–100 μm de diamètre qui ont été recristallisés à 200–400 μm. Les ciments dolomitiques sont représentés par une dolomie brute qui remplie les vacuoles (100–200 μm) et une dolomie en forme de dos d'âne (1000 μm). Toutes les quatre générations de dolomie ont donné des valeurs de $\delta^{18}O$ similaires de –10,7 à –16,5 ‰ (PDB), des valeurs $\delta^{13}C$ de +0,7 à +3,2 ‰ (PDB) et des rapports de Sr isotopique de 0,70826 à 0,70846 qui ne concordent pas avec les valeurs attendues du Dévonien moyen pour les carbonates ou l'eau de mer. Nous considérons que ces données indiquent que des mélanges de fluides de bassins préLaramide, chauffés par enfouissement au cours de l'orogenèse du Laramide, étaient présents durant le Laramide tardif lorsque les dolomies ont recristallisé et (ou) ont précipité avant la migration du pétrole et de l'accumulation du réservoir Rainbow « A ». Ainsi, l'utilisation combinée du paléomagnétisme, de la géochimie et de la pétrographie s'est avérée une technique utile pour la datation et l'identification des événements de dolomitisation et des chemins de migration des hydrocarbures.

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Introduction

This study was undertaken for two reasons. One was to test the paleomagnetic method for orienting drill core, given the unusual circumstance of having an horizontal length of core from a reservoir. The second was to petrographically and geochemically quantify and date dolomitization and hydrocarbon migration events in the Husky Rainbow "A Pool" reservoir carbonates of the Western Canada Sedimentary Basin (WCSB) (Fig. 1), and to try to relate these events to fluid flow events in the WCSB.

Structural and fabric analysis of cores is an essential part of most hydrocarbon exploration projects. It is usually done on unoriented cores, although much more information can be obtained from azimuthally oriented cores. This is because traditional methods of orientation are costly, orienting equipment is expensive, the drilling rate is decreased to permit measurement, and the decision must be made before the core is examined in detail to determine the value of orienting it. Nelson et al. (1987) have reviewed the various methods and assessed their accuracy.

Paleomagnetic studies are normally done on oriented samples from outcrops. Recently, however, it has been recognized that paleomagnetic methods can provide several types of valuable information for hydrocarbon exploration (Turner and Turner 1995). One of its uses is to orient drill core into its in situ geographic coordinates after it has been in storage for several years to determine, for example, the orientation of a reservoir's fracture system. The orientation can be done paleomagnetically in two ways.

The common method is to use the viscous remanent magnetization (VRM). This is done by measuring the lower temperature or lower coercivity VRM component in the core and then aligning its direction to the present Earth's dipole magnetic field (PEMF) direction using the well's orientation as the rotational axis (Van der Voo and Watts 1978). A modern VRM is usually preserved in petroliferous carbonate strata but it often requires very sensitive paleomagnetic instrumentation to measure it. Most researchers accept the use of the PEMF direction for a reference (e.g., Van der Voo and Watts 1978; Kodama 1984), however Rolph et al. (1995) have recently suggested using the geographic or time-averaged Earth's dipole magnetic field (GEMF). They argue that 730 000 years of spontaneous realignment since the last major polarity reversal will be preserved in the VRM. While this is true, the acquisition or decay of VRM is logarithmic with time, so that a short time period in the recent past may still represent a much larger fraction of the VRM signal. To date, we have found that the PEMF provides a better reference (Lewchuk et al. 1998).

Most cores also preserve a higher temperature or higher coercivity characteristic remanent magnetization (ChRM) component. Typically it has been ignored in VRM core-orientation studies because it is time consuming, often more difficult to isolate, and it has less direct utility. Nevertheless, if the core can be oriented by its VRM or by some other method, then the ChRM can be compared to the cratonic apparent polar wander path (APWP) to date its time of acquisition (e.g., Perroud et al. 1995; Lewchuk et al. 1998). In this way, for example, unfossiliferous strata with a primary ChRM can be dated or, alternatively, a secondary diagenetic event such as dolomitization, which remagnetizes the strata with a secondary ChRM, can be dated. Such remagnetization events have been causally related to massive regional fluid flow and hydrocarbon migration events (McCabe and Elmore 1989; Oliver 1986, 1992), and they can be expected to cause significant modifications in the geochemical systematics of the strata (Elmore et al. 1993; Burton et al. 1993).

If core is available from three or more wells with substantially different orientations, then it is possible to use the measured ChRM both to date its time of acquisition and to orient the cores. The "three-well ChRM" method of core orientation uses a triangulation technique similar to that used to determine the bedding or fabric dip direction from unoriented well cores (see p. 161, Ramsay and Huber 1989; Versteeg and Morris 1994). This is done by determining the mean inclination of the ChRM for each well relative to the well's axis. Rotating the inclination about the axis generates a small circle on a stereonet, and the three small circles from the three wells will intersect in the only geographic orientation for the ChRM that is permissible within statistical error. Each core can then be oriented by matching its measured ChRM to the calculated vector orientation.

Geology

The Rainbow petroleum field is located about 640 km northwest of Edmonton, Alberta (Fig. 1), within the Middle Devonian (Givetian) Upper Elk Point Subgroup. This subgroup ranges in thickness up to about 330 m thick in the central part of the Rainbow sub-basin, and it onlaps the Peace River Arch to the south (cf. Qing and Mountjoy 1989; Frydl 1989). Stratigraphically, it consists of the following formations in ascending order: the dolomitized, micritic, and reefal Keg River Formation; the evaporitic and evaporitic-carbonate Muskeg Formation; the carbonate-dominated Sulphur Point Formation; and the clastic-dominated Watt Mountain Formation (Schmidt et al. 1985; Qing and Mountjoy 1989). The field consists of several pools in stratigraphic traps formed by carbonate pinnacle-reef buildups of the Upper Keg River Member that are enclosed by evaporites and bituminous limestones of the overlying Muskeg Formation (Schmidt et al. 1985). The Rainbow "A" pool is the oldest pool in the Rainbow area with ca. 1.32 billion barrels of oil in place.

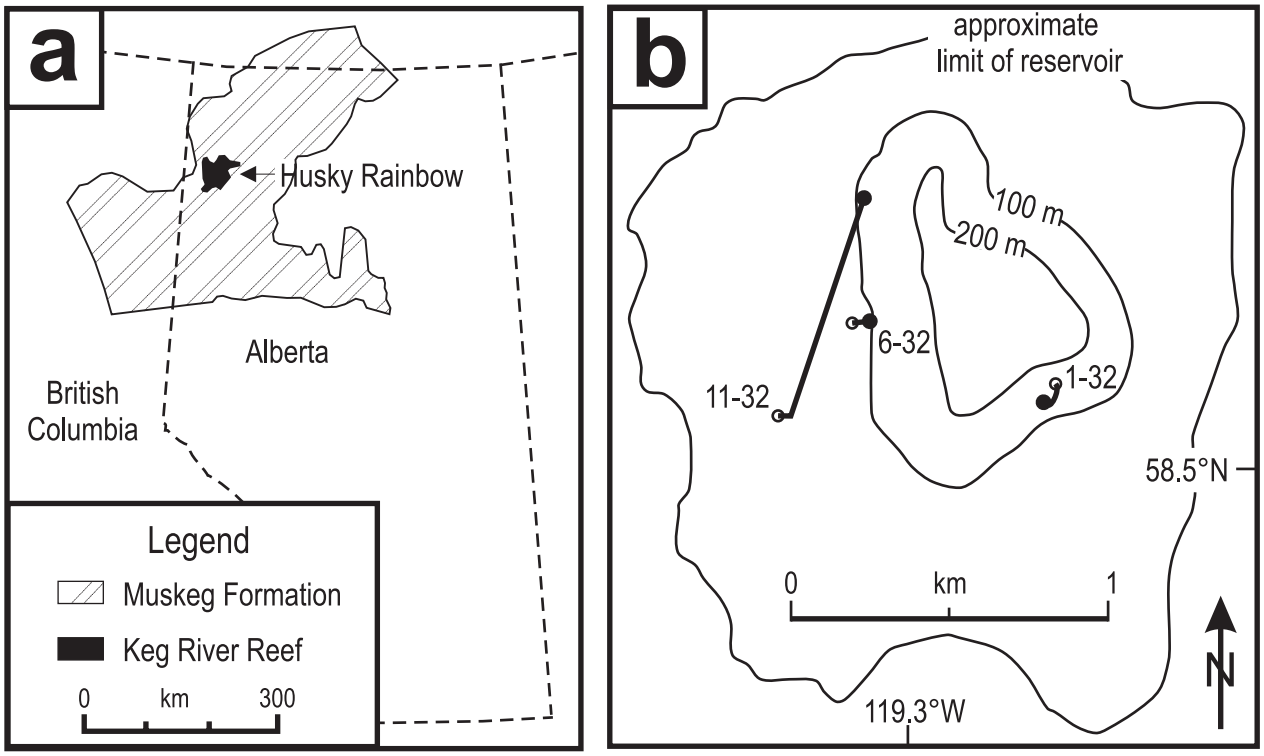
Methods

Sampling

The samples for paleomagnetism, petrography, and geochemistry come from three wells in the "A Pool" of the Husky Rainbow Field in northwestern Alberta (Fig. 1*a*). Where sampled, well 1 deviated from vertical by 14.5° towards an azimuth of 244° (universal well locator = 01–32–109–8W6); well 6 was vertical (06–32–109–8W6); and, well 11 was horizontal towards an azimuth of 019° (11–32–109–8W6) (Fig. 1*b*). The samples were distributed along 1633 m, 14 m, and 230 m of the well cores, respectively. The cores provide no evidence of primary dip in this reservoir. We chose this reservoir because it has unoriented core available

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Fig. 1. (*a*) Map of Western Canada Sedimentary Basin showing the Middle Devonian (Givetian) Muskeg Formation and the location of the Upper Keg River Member pinnacle reef of the Husky Rainbow field within it. (*b*) Isopach map of the "A Pool" reservoir and the locations of the three wells. The top of each well is shown by an open circle, the bottom of each well by a solid circle.



from vertical, inclined, and horizontal holes. Thus it provided the opportunity to compare the VRM and three-well ChRM methods of core orientation and to date dolomitization or dolomite recrystallization in the reservoir.

Hailwood and Ding (1995) and Audunsson and Levi (1997) have described paleomagnetic sampling techniques for well core, and our procedure has been described in more detail in Lewchuk et al. (1998). Ideally, a core should first be reassembled into long continuous segments to permit direct comparison of the specimens' remanence directions within the segment. Unfortunately, the Husky Rainbow cores proved to be extensively broken and jumbled from prior investigations. In addition numerous preexisting saw cuts, made perpendicular to each core's axis for whole-core porosity and permeability measurements, severely hampered this effort. Thus it was not possible to reassemble segments. The paucity of virgin core, in addition to industry and government restrictions on minimum sample spacing, meant that we were severely limited in the amount of material available for the study. We were allowed to drill only 10 plugs from each well. Each plug yielded from one to three specimens each, resulting in a total of 64 specimens for the paleomagnetic collection. The cores did have a preexisting master orientation line (MOL) or scribe line so we measured the angle between each plug azimuth and the MOL. Even though its true azimuth is rarely known, a MOL can also be used for the relative comparison of remanence directions in a given core. Unfortunately the MOL clearly had not remained straight during drilling. Thus, the declination data for our entire collection was arbitrary and could not be used in the remainder of the analysis. In addition, we wanted to avoid slicing specimens from the perimeter of the core because of the potential existence of drilling-induced isothermal remanent magnetization at the perimeter that is acquired during initial boring of the core at the well site (Pinto and McWilliams 1990). However, the Husky Rainbow core has a relatively small diameter of ~8 cm, so this was not feasible. Instead, we noted which specimens were close to the edge and later compared their results to those from the core's interior.

In most cases the samples for petrologic and geochemical analysis were taken from the same pieces of core as the paleomagnetic plugs.

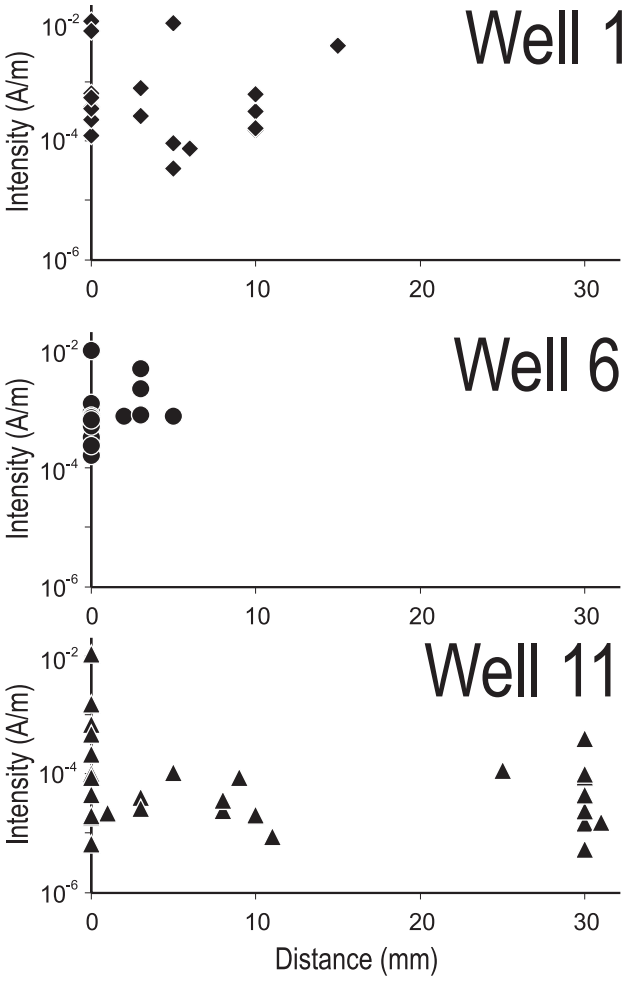
Petrology and geochemistry

Thin sections from the three wells were prepared and examined by applying standard petrographic microscopy and cathodoluminescence (CL) microscopy using a Technosyn ™ cold cathodolominescence stage with a 12–15 kV beam and a current intensity of 0.42–0.43 mA. In addition, fluorescence characteristics were examined with a Nikon EPI Fluorescence ™ stage connected to a petrographic microscope.

Oxygen and carbon isotopes were extracted from dolomite samples (n = 39) using a microscope-mounted drill assembly. The samples were reacted in vacuo with 100% pure phosphoric acid for at least 4 h at 50°C for dolomite using the method described by Al-Aasm et al. (1990). The evolved CO_2 gas was analyzed for isotopic ratios on an SIRA-12™ mass spectrometer. Values of O and C isotopes are reported

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Fig. 2. NRM intensity vs. distance from the edge of core for all specimens studied regardless of whether it was a top or bottom edge.



in per mil (‰) relative to the PDB standard and were corrected for phosphoric acid fractionation. Precision was better than 0.05‰ for both $\delta^{18}O$ and $\delta^{13}C$. The $^{87}Sr/^{86}Sr$ ratios of selected dolomite samples (n = 8) were determined after washing the samples with distilled water to remove the pore salts that result from drying. The dolomite samples were then reacted with 0.1% HCl and analyzed using an automated Finnigan MAT 261™ mass spectrometer equipped with nine Faraday collectors. All analyses were performed in the static multicollector mode using Re filaments. Correction for isotope fractionation during the analyses was made by normalization to $^{86}Sr/^{88}Sr = 0.1194$. The mean standard error of mass spectrometer performance was 0.00003 for NBS-987.

Four doubly polished samples were used for fluid inclusion analysis on a Linkham TH 600 heating-freezing stage. Homogenization temperatures (Th) and salinity values (Tm) were measured for each inclusion. Salinities were calculated from final ice melting temperatures using the equation of Potter et al. (1978) in terms of H_2O–NaCl system.

Natural remanence magnetization

The prepared specimens were stored in a steel-shielded room with an ambient magnetic field of ~0.2% of the Earth's field intensity for eight months to allow their most unstable VRM components to decay. Such components are acquired by a specimen during storage and preparation, and are of no geologic interest. All subsequent measurements were done without removing the specimens from the room.

First the natural remanent magnetization (NRM) of each specimen was measured on an automated Canadian Thin Films CTF-420™ cryogenic magnetometer. It can reliably measure a remanence down to ~4×10^{-6} A/m, below which the contributions of paramagnetic and diamagnetic minerals and of spurious noise give erratic directions. The median NRM intensities were 3.5, 7.8, and 0.45×10^{-4} A/m for wells 1, 6, and 11, respectively. Plots of NRM intensity versus distance from the cores' perimeters (Fig. 2) suggest that there may be a slight decrease away from the perimeter, but that it is neither statistically significant nor comparable to the order of magnitude decrease seen by Pinto and McWilliams (1990). Therefore, although the possibility of a drilling-induced remanence cannot be entirely eliminated, it does not appear to be a problem in these cores.

Alternating field demagnetization

Sixteen specimens were alternating field (AF) demagnetized in 25 steps up to a peak field of 150 mT using a Sapphire Instruments SI-4 ™ AF demagnetizer. Most of these specimens showed near-linear decay to the origin of a vector component plot, losing >50% of the NRM intensity by ~20 mT (Fig. 3). The linearity from 0 to 20 mT indicates that either negligible VRM is present or that AF demagnetization is ineffective in separating the VRM and ChRM components. This problem is discussed further by Lewchuk et al. (1998).

Isothermal remanent magnetization

Twelve of the 16 AF demagnetized specimens were subjected to isothermal remanent magnetization (IRM) analysis to more fully characterize their magnetic mineralogy. Using a Sapphire Instruments SI-6 ™ DC pulse magnetizer, they were magnetized in 11 steps up to 900 mT and then AF demagnetized in 7 steps to 70 mT. All of the specimens exhibit a rapid acquisition of IRM, reaching saturation by 300 mT (Fig. 4). This indicates that magnetite or possibly pyrrhotite may carry the NRM, but that neither hematite or goethite are even minor contributors because they acquire IRM much more slowly and are not saturated by 900 mT. The IRM intensity decay curves on AF demagnetization show by their rate and form that single domain to pseudosingle domain magnetite or low-titanium titanomagnetite is the likely ChRM carrier. No difference in IRM behavior was found between cores from the three wells.

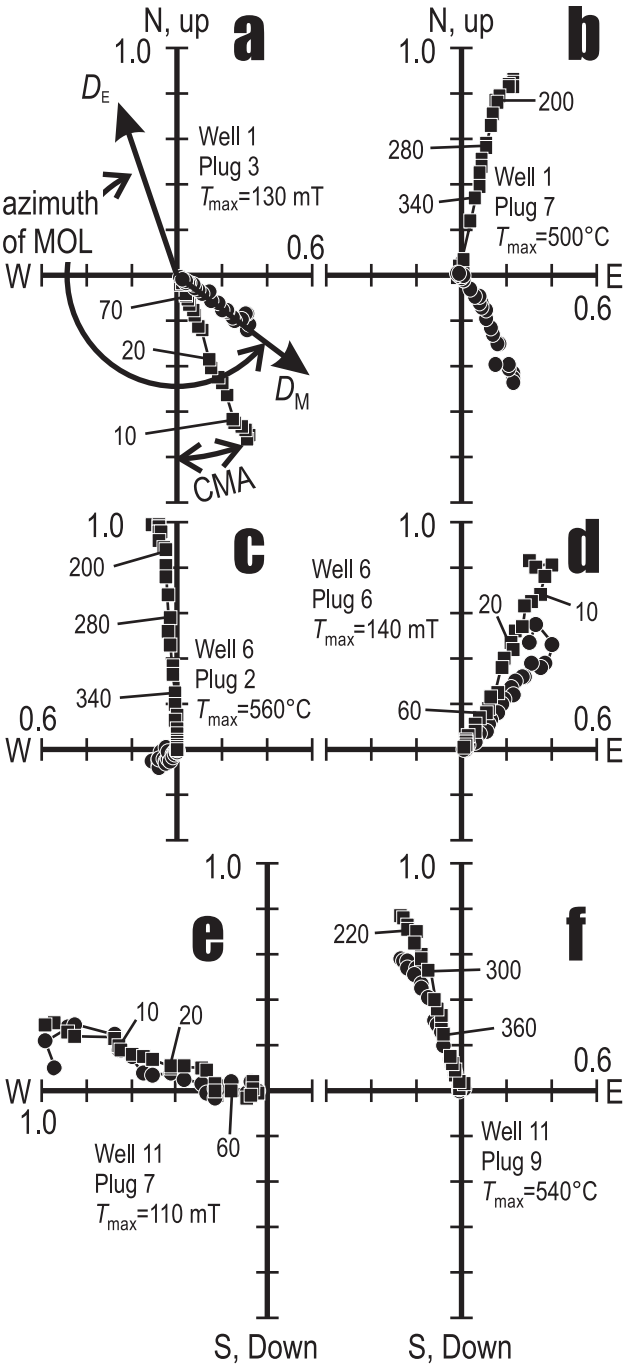
Thermal demagnetization

The remaining 48 specimens from the collection were thermally demagnetized in 28 steps to a 570°C peak temperature using a Magnetic Measurements MMTD-1™ furnace. Noting that the core was at an ambient temperature of ~85°C in the Husky Rainbow reservoir, and considering the time-unblocking temperature relationship for magnetite (Pullaiah et al. 1975), all data from steps up to ~175 ± 25°C were rejected, because this remanence would have been blocked into the core when it was being removed from the hole.

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Again nearly all specimens behaved similarly. They showed a modest remanence intensity decrease below 250°C, the range in which the VRM component would be expected to reside, indicating that the VRM method for orienting the core would probably not work, and this proved to be the case. Above ~250°C, most specimens yielded a remarkably linear decay to the origin and thus a well-defined ChRM (Fig. 3). In addition, none of the specimens showed an accelerated rate of decrease in the diagnostic 275–325°C unblocking temperature range of pyrrhotite, confirming that it is not a contributor to the ChRM.

Fig. 3. Normalized orthogonal demagnetization plots for six representative specimens of the collection. Circles are projections in the horizontal plane and squares are projections in the vertical plane. T_{max} is the maximum demagnetization treatment for that specimen and the step highlighted in white shows either the 200°C or 20 mT step as appropriate. All data are presented after correction for plug azimuth and inclination, assuming that the borehole axis is vertical and the arbitrary master orientation line (MOL) points north. By plotting the data in this manner, the true core magnetization angle (CMA) is the plotted inclination.



Petrography and geochemistry of dolomite

The Keg River carbonate in Rainbow "A" Field has been extensively dolomitized with the primary limestone components severely obliterated. The main type of dolomite observed in all cores is fabric-destructive, subhedral to anhedral matrix dolomite (60–100 μm; Fig. 5*a*). This dolomite replaces the original fossil components and surrounding matrix. Faint relics of crinoids, corals, and other allochems are sometimes visible. This dolomite is sometimes porous with typical intercrystalline porosity (Fig. 5*b*). Under CL this dolomite is showing a dull-red core with a thin, bright red rim. The contact between the core and the rim is very irregular. Matrix dolomite is characterized by a narrow range of depleted δ^{18} O values (range between –10.7 and –14.8 ‰ PDB; Fig. 6) relative to the values postulated for Middle Devonian carbonates (Hurley and Lohmann 1989) and positive δ^{13}C values (+0.9 to 3.0 ‰ PDB). The mostly depleted δ^{18} O values are those of porous matrix dolomite.

The second type of dolomite is slightly coarser than subhedral matrix dolomite (200–400 μm; Fig 5*c*). Dolomite crystals are anhedral with nonplanar crystal boundaries. It occurs also as replacement, and it may represent recrystallized earlier-formed matrix dolomite. These dolomite crystals show some fractures. Their oxygen and carbon isotopic values (δ^{18} O = –11.6 to –11.7; δ^{13}C = +2.8 to +2.9 ‰) are very close to those of subhedral matrix dolomite. These dolomites show similar CL characteristics to fine matrix dolomite above.

Also present is a third type of dolomite. It is a vug-rimming, porous, coarse dolomite cement (100–200 μm; Fig. 7*a*) that infills vugs, fossil pores, and fractures. Its CL images are characterized by having a dull-red core and a thin, bright-red rim with a sharp contact between these two zones. The δ^{18} O values range from –11.7 to –14.3‰, and δ^{13}C values range from +0.7 to +2.9‰ (Fig. 6).

The last or fourth generation of dolomite in the studied cores is saddle dolomite (1000 μm, Fig. 7*b*), which occurs as cement in voids and fossil chambers as well as fracture-fill. Under CL, it is characterized by dull-red color and an irregular, thin, bright-red rim. Its oxygen and carbon isotopic values slightly overlap the other generations of dolomite with δ^{18} O ranging between –11.5 and –16.5‰ and δ^{13}C values ranging between +2.3 and +3.2‰ (Fig. 6). The saddle dolomite is crosscut by large amplitude stylolites that are filled with bitumen. Secondary anhydrite occurs sometimes as void–filling cement and postdates saddle dolomite.

Both matrix dolomite and saddle dolomite show comparable δ^{18} O values and $^{87}Sr/^{86}Sr$ ratios (Fig. 8), and both some-

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Fig. 4. Normalized isothermal remanent magnetization (IRM) plots showing (*a*) acquisition of the magnetization from a DC field to 900 mT and (*b*) decay of the IRM at 900 mT from AF demagnetization to 70 mT. Diamonds, circles and triangles show specimens from wells 1, 6, and 11, respectively. SD, PSD, and MD are the boundaries for single, pseudosingle and multidomain magnetite, respectively.

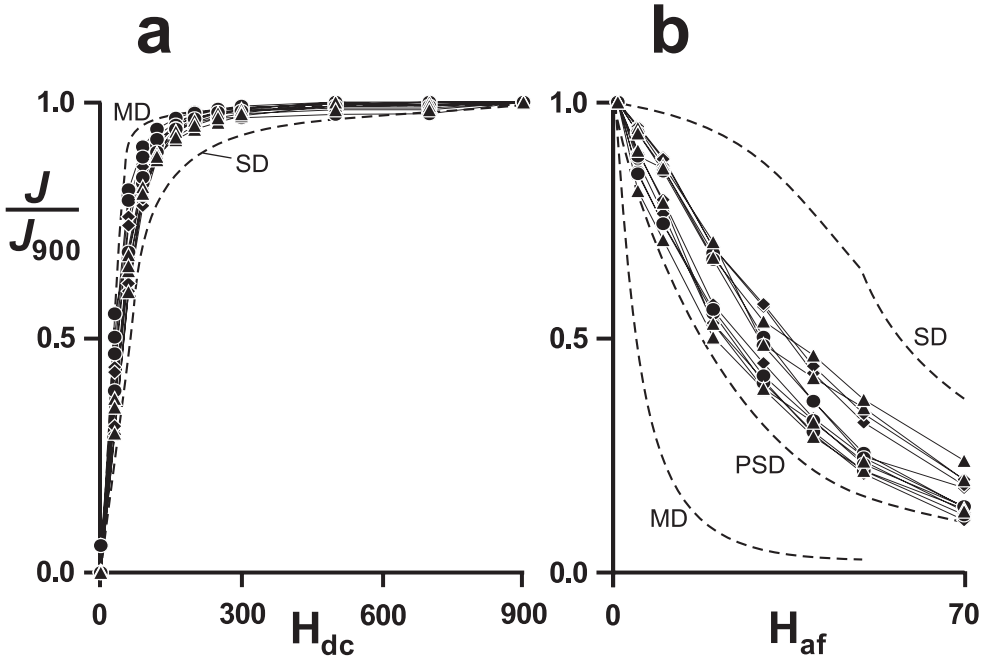


Table 1. Summary of fluid-inclusion data for diagenetic phases.

Diagenetic mineral phase	*T*h ± S.D. (°C)	*T*m ± S.D. (°C)	Salinity (wt.% NaCl)
Anhydrite (n = 5)	119.7 ± 2.6	–18.1 ± 0.7	21.3 ± 1.0
Subhedral–anhedral dolomite (n = 11)	102.9 ± 1.4	–8.3 ± 0.5	12.0 ± 0.8
Saddle dolomite-cores (n = 3)	100.9 ± 1.8	–27.3 ± 1.4	27.0 ± 1.4
Saddle dolomite-rims (n = 4)	100.1 ± 0.5	–9.3 ± 0.4	13.1 ± 0.8

what deviate from the postulated values of Middle Devonian carbonates and seawater (Denison et al. 1997).

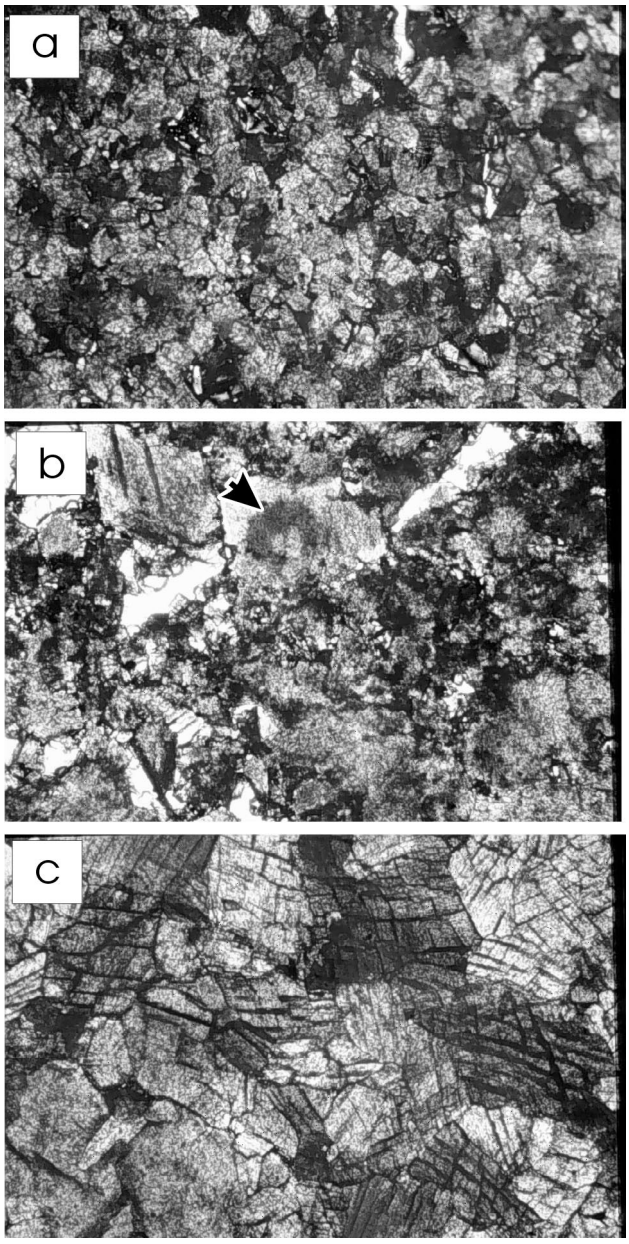
Primary, liquid-vapour fluid inclusions were observed in matrix dolomite, saddle dolomite, and anhydrite (Table 1). Table 1 shows homogenization temperatures, melting temperatures, and corresponding salinity values for fluid inclusions in these phases. The average homogenization temperature (*T*h) for coarse, recrystallized?, subhedral to anhedral matrix dolomite is ca. 103°C and measured melting temperature (*T*m) is –8.3, which translate to salinity value of ca. 12 wt.% equivalent NaCl. The homogenization temperature and salinity value for saddle dolomite rims (*T*h = 101°C; *T*m = –9.3°C) are very comparable to that of matrix dolomite. However, the salinity value for saddle dolomite core (*T*m = –27.3°C) are different from the rims. The homogenization temperature for late anhydrite is ca. 120°C and *T*m = –18°C, which translates to salinity value of ca. 21 wt.% equivalent NaCl.

Core magnetization angles

The directions of the magnetization components in each specimen were identified on orthogonal vector component plots (As and Zijderveld 1958) and calculated using the least-squares fitting technique (Kirschvink 1980). Finding that the VRM method of orientation was not possible, the three-well ChRM method became the only possible option. For it, the core magnetization angle (CMA) is used. It is the deviation angle of the specimen's magnetization from the core axis which is the same as the complement of the measured remanence inclination. Shown on Fig. 9 are the specimen CMA values determined for the three wells. Some points need to be made. One is that plugs 3, 4, and 10 in well 1 gave apparently normal or downward ChRM directions. Two possibilities exist. First, and more likely, the directions are truly bipolar, recording reversed and normal epochs of the Earth's magnetic field. Second, the short pieces of core from which plugs 3, 4, and 10 were taken had been put back upside down in the core box during previous examinations. Also, for a horizontal bore core without continuous segments, polarity of an individual segment can be ambiguous when the inclinations are shallow. Regardless, it is still valid to use the absolute values of inclination to obtain a CMA. A second observation is that there is scatter in the data. Therefore, we opted for a conservative two-tiered statistical approach to get the core mean CMA. First, we averaged the specimen CMA values to get an arithmetic mean CMA for each plug. In well 1, the paired specimens for

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Fig. 5. (*a*) Photomicrograph of porous, fabric-destructive, subhedral matrix dolomite replacing the original mud matrix and fossils (1–32–109–8W6, 1927 m; field of view is 2.4 mm). (*b*) Photomicrograph of porous, matrix dolomite with dolomitized crinoid ossicles (arrow). (11–32–109–8W6, 2346 m; field of view is 2.4 mm). (*c*) very coarse, fractured, anhedral, fabric-destructive, matrix dolomite (6–32–109–8W6; 1914 m; field of view is 2.4 mm). All photographs were taken under crossed



plugs 3 to 7 and 10 all gave consistent and similar values with a mean CMA of –71.1° (k = 123, α_{95} = 5.9°) using the McFadden and Reid (1982) method for calculating the unbiased confidence limits from inclination-only remanence directions. Based on this result, the CMA values for the remaining specimens were accepted only if they fell within two times the α_{95} value of the initial mean CMA. Thus, in total, 15 of the 20 specimens from 9 of the 10 plugs gave accepted values. The plug mean CMAs were then averaged to

Fig. 6. Carbon and oxygen isotopic compositions of dolomite types. See text for explanation.

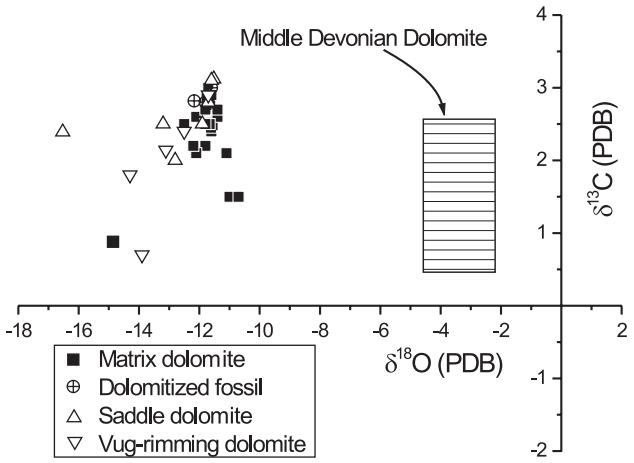
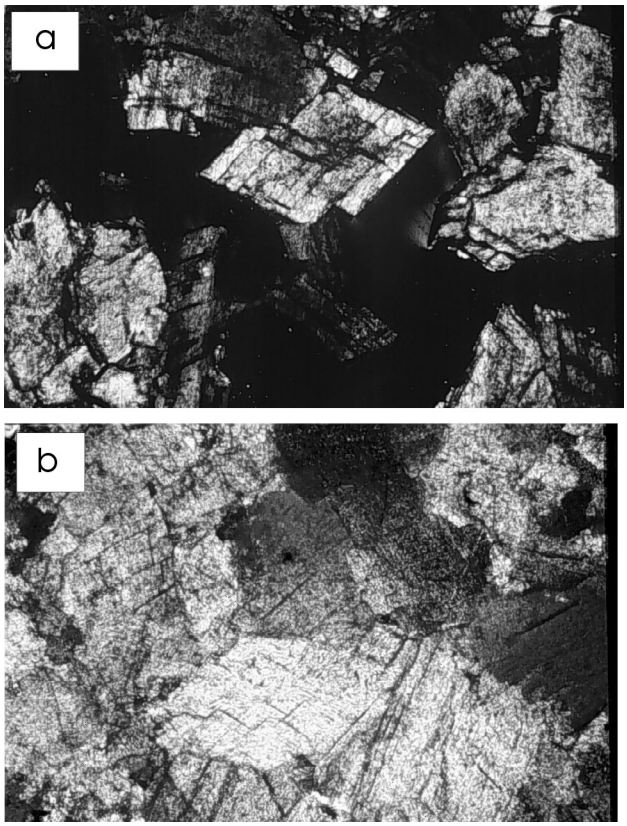


Fig. 7. (*a*) Photomicrograph of coarse, euhedral, vug-rimming dolomite. Note the inclusion–rich core (6–32–10–8W6; 1914 m; field of view is 1.3 mm). (*b*) Photomicrograph of vein-fill saddle dolomite cement (1–32–109–8W6, 1935.5 m; field of view is 2.4 mm). All photographs were taken under crossed polars.



get the arithmetic core mean CMA, following the McFadden and Reid (1982) of –69.7° (k = 113, α_{95} = 5.0°). For well 6, all 15 measured specimens gave coherent ChRM directions or values. Following the same process as for well 1 resulted in the rejection of one specimen CMA value from plug 6, with the resulting mean CMA for all 10 plugs of –73.3° (k = 33.9, α_{95} = 9.0°). Eight plugs from well 11 had two or three

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Fig. 8. The $\delta^{18}O$ vs. $^{87}Sr/^{86}Sr$ compositions of matrix dolomite and saddle dolomite samples. See text for explanation.

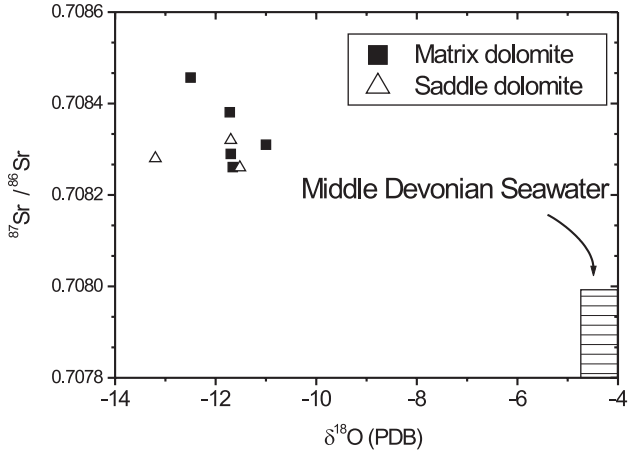
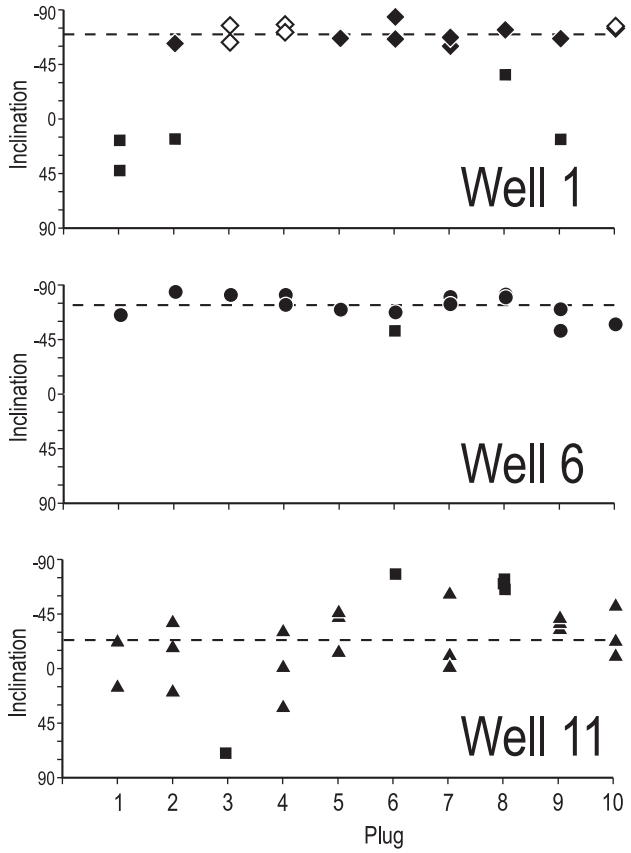
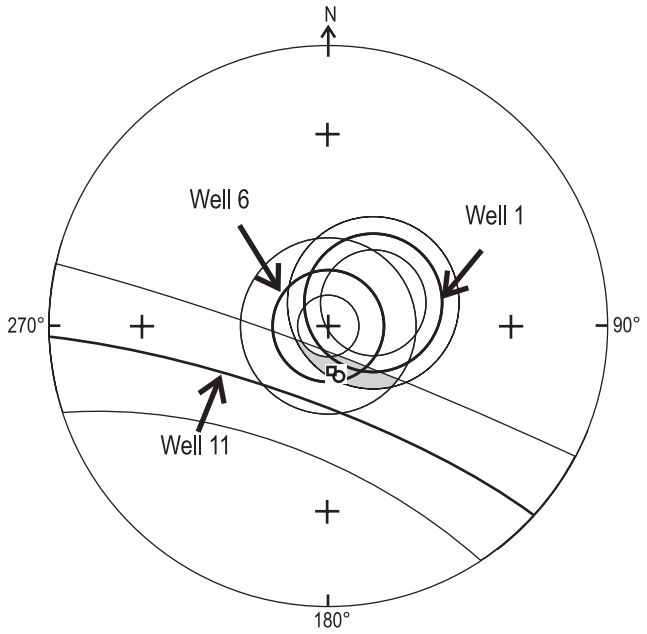


Fig. 9. Observed core magnetization angles for each of the ten plugs in each of the three wells. Diamonds indicate accepted data points. Squares indicate those data points that were rejected from the final calculation. Open symbols indicate reversed and probably antipodal angles.



specimen CMA values which were averaged, and the plug averages were in turn averaged to get an initial core mean CMA. Using two times its α_{95} value led to the rejection of data from single-specimen plugs 3 and 6 and multiple-specimen plug 8. The remaining seven plugs with data from 20 specimens gave a core mean CMA of –23.8° (k = 18.0, α_{95} = 15.4°). The increased scatter in this result relative to

Fig. 10. Equal-area stereonet projection of the mean core magnetization angles or CMA's (thick lines) about all three well azimuths with their bands of 95% confidence (thin lines). The common region of 95% confidence is shaded. All data are plotted in the upper hemisphere. The open circle is the unit mean ChRM direction by Method 4 in Table 2, the open square for Method 5.



wells 1 and 6 is attributed to the weaker initial NRM intensities in specimens from well 11, resulting in a majority of the ChRM being isolated in the 10^{-6} A/m range or relatively close to the magnetometer's sensitivity limit.

Aliasing from the overlap of the remanence directions' population over the core axis or 90° position of a data set will result in undetectable errors unless a very large data set is available (McFadden and Reid 1982). For example, an inclination of 88° on the near side of vertical cannot be distinguished from one of 88° on the far side, because no declination is available. Several different techniques for calculating the amount of the bias and the error limits have been proposed (e.g., Briden and Ward 1966; Kono 1980; McFadden and Reid 1982; Cox and Gordon 1984; Enkin and Watson 1996; Westphal and Gurevitch 1996), but no consensus has yet developed in the paleomagnetic community as to which is preferable. Our conservative method for handling outlier plug CMA values, described above, meant that the core mean CMA values were greater than the amount by which any of the individual plug values deviated from their mean CMA values, thus satisfying the McFadden and Reid (1982) criteria and resulting in the acceptance of the arithmetic mean CMAs as being accurate to within <1°. Following the numerical integration method of Enkin and Watson (1996) gives CMA's of well 1, –70.2° $\pm^{4.0°}_{5.2°}$ (±1.96 × standard deviation or ~ a_{95}); well 6, –76.5° $\pm^{6.3°}_{10.1°}$; and well 11, –24.2° $\pm^{10.8°}_{12.6°}$. Note that the difference in the CMAs determined by the two methods are <1°, except for well 6 where it is 3.2°. Plotted on Fig. 10 are the three small circles generated from rotating the core arithmetic mean CMAs about the borehole or core axes along with their 95% confidence limits. From Fig. 10, it is obvious that the unit mean

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direction must be in the southeast up quadrant, where all three regions of 95% confidence overlap.

Unit mean ChRM direction

The problem of finding the unit mean ChRM direction from three core mean CMAs with different variances from wells drilled at different angles does not appear to have been addressed before in the literature. Several ad hoc approaches are considered (Table 2).

A very conservative first approach is to deem each well to be considered equal to one paleomagnetic site and then determine the mean direction of the three closest points of intersection for the three pairs of CMA small-circle cones of revolution (Method 1, Table 2). This results in a very large $\alpha_{95,}$ because only three values are used. Noting that each well is represented by 15 to 20 specimens from 7 to 10 plugs along 14–1633 m of bore core, it would be more reasonable to deem each well to be equivalent to two conventional paleomagnetic sites to calculate the α_{95} value (Method 2, Table 2). Note, however, that neither of these methods utilizes the confidence limits determined for each CMA value.

A third approach is to argue that the two small circles given by the CMA cones for wells 1 and 6 intersect in two places, and that the well 11 small circle should not be used, because it does not intersect either of the other two small circles, but that it does indicate that the "southeast and up" intersection is the correct result (Fig. 10). The α_{95} value is then obtained using the root mean square of the two well values (Method 3, Table 2). There is, however, no a priori reason why the CMA small circles must intersect for the results to be valid. Thus, a fourth method for finding the unit mean ChRM direction is to find the best-fit point for each pair of wells in turn. For wells 1 and 6, this is simply the intersection of the two small circles. For wells 6 and 11 and 1 and 11, the probability (p) is varied in the McFadden and Reid (1982) equation until a common point of equal probability was found for the limits about the two mean CMAs on the great circle of the plane containing the axes of both holes. The centre of the spherical triangle of the three best-fit points is then taken as the optimum ChRM direction. Recalculating the measured plug mean CMA direction with respect to vertical for wells 1 and 11, all available 26 values were then rerun through the McFadden and Reid (1982) equation to calculate the confidence limit (Method 4, Table 2). It gives a pole position for the Husky Rainbow "A" reservoir of 164°E, 84°N (A_{95} = 10°) (longitude, latitude (radius of cone of 95% confidence)).

The fifth approach, suggested and calculated by R.J. Enkin (personal communication, 1999) is to adapt McFadden and McElhinny's (1988) method for calculating the best-fit mean of intersecting great circles to intersecting small circles. Using a weighting function proportional to $1/(\alpha_{95} + 5°)^2$ to obtain the confidence limits of the small circles derived by numerical integration, an optimum fit ChRM direction is calculated. Its confidence limit is then obtained by parametric resampling of the 95% confidence intervals of the small circles a total of 250 times, assuming they have a symmetric Gaussian distribution. For each resampling, the mean ChRM direction was with 1° of the optimum fit value (Method 5, Table 2). This method gives a pole position for the reservoir dolomite of 191°E, 84°N, A_{95} = 12°.

Table 2. Unit mean direction based on differing calculation methods.

	I			II		
Method	Dec	Inc	α95	Dec	Inc	α95
1	170	–73.7	15.6	175	–72.1	12.3
2	170	–73.7	7.6	175	–72.1	6
3	142	–73.3	7.3	149	–73.8	8.9
4	168	–73.5	5.8			
5				169	–75.3	6.7

Notes: I is using core mean angles (CMA's) calculated using McFadden and Reid (1982). II is using Enkin and Watson (1996). Dec, Inc, α95 are the declination, inclination and radius of the cone of 95% confidence, respectively. All values are given in degrees. Methods 1 to 5 are described in the text.

Discussion

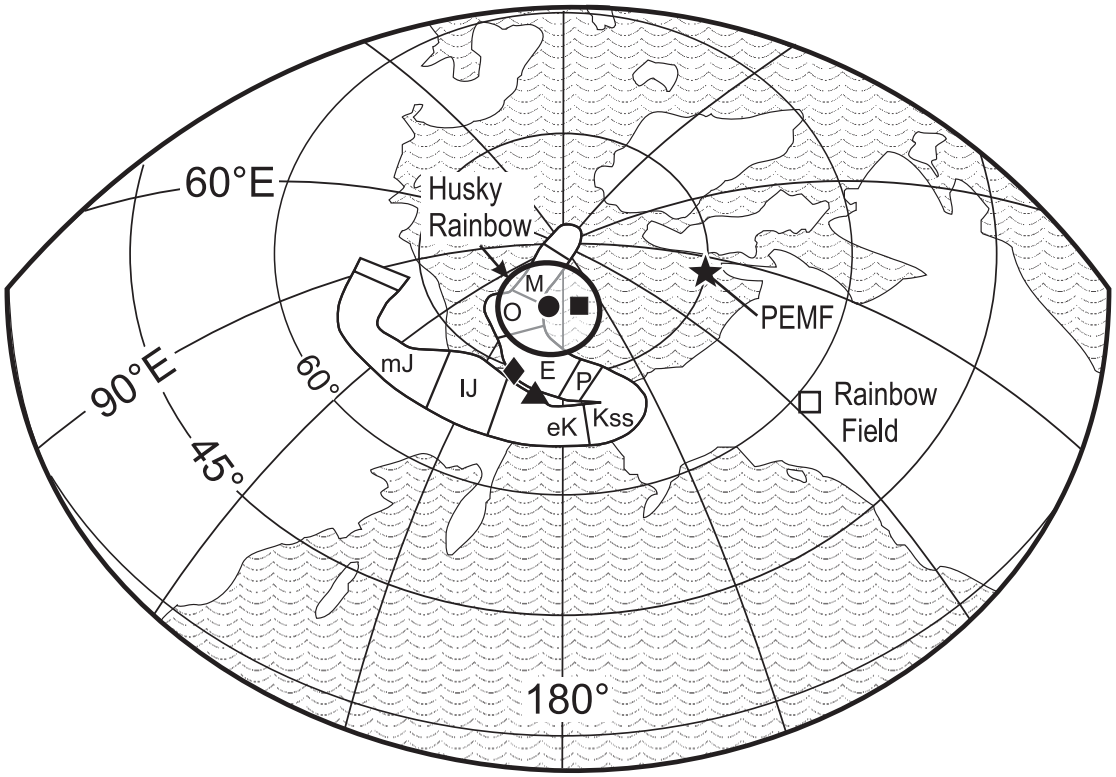
Paleomagnetism

The pole positions derived from Methods 4 and 5 (Table 2) at 164°E, 84°N, A_{63} = 6° and at 191°E, 84°N, A_{63} = 7°, respectively, are only 3° apart and are not statistically significantly different at 95% confidence (Fig. 11). The 3° difference stems mostly from the latter method invoking a larger correction for aliasing. Given that we know of no tectonic or other reason to cause the pole to fall off the APWP, we consider the Method 4 to be preferable. Method 4 also provides a "middle of the cluster" pole relative to other methods. Its cone of 95% confidence covers parts of the Paleocene to Miocene portion with the Middle Eocene to Middle Miocene portion covered by the circular standard deviation cone on APWP for the North American craton from Besse and Courtillot (1991) (Fig. 11). Thus any preexisting primary Devonian remanence has been entirely overprinted so that the ChRM dates dolomitization or recrystallization of dolomite in the reservoir at ~40 ± 25 Ma, indicating that it was the product of a late Laramide orogenic event. This age then limits the timing of charge in this oil reservoir to be after this date.

The failure of the specimens to retain an isolatable VRM means that we were unable to compare the relative precisions of the VRM and three-well ChRM methods for orienting bore core. Obviously, however, the latter method proved successful, despite having weak NRM intensities and unfavourable sampling conditions (i.e., relatively few specimens, lack of an MOL, broken and cut core that could not be reassembled into long segments, potentially inverted core, etc.). A reasonably accurate age would have been obtained had the specimens from only the vertical well 6 and the slightly-inclined (i.e., 14.5° off vertical) well 1 been used. With data from these two wells only, two optimum fit ChRM points are possible. They are where the two small circles intersect in the northwestern up and southeastern up quadrants (Fig. 10). The pole position for the northwestern direction is in present-day northern Mexico, which is an impossible result, being >50° of arc away from any part of the Paleozoic to present APWP for North America. Conversely, the pole for the southeastern direction at 169°E, 71°N or at 169°E, 74°N, depending on the method of CMA calculation, falls on the Cretaceous to Eocene portion and has a cone of 95% confidence that overlaps both the Early Cretaceous and

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Fig. 11. Apparent polar wander path for stable North America after Besse and Courtillot (1991) from the Middle Jurassic (mJ) to Miocene (M) epochs. Kss is the Cretaceous magnetic stillstand. The solid circle is the determined pole using the preferred Method 4 (Table 2), as described in the text, with its circular standard deviation cone (thick line). For comparison the square is the pole based on Method 5. The poles for both variants of Method 1 and 2 similarly fall inside the cone. The triangle and diamond show the poles for the two variants of Method 3, for which only the data from wells 1 and 6 were used. The star is the Earth's present magnetic field pole (PEMF).



Eocene portions of the APWP (Fig. 11). Thus, it defines the obvious choice for a best-fit ChRM direction, had only the two cores been available, and it gives a sufficiently precise result to enable stored core to be oriented azimuthally to within ~10°.

The finding of a late Laramide ChRM direction for the Husky Rainbow "A Pool" reservoir is consistent with other recent paleomagnetic results from Paleozoic carbonates in the WCSB. These Laramide (i.e., Late Jurassic to Middle Eocene) results have been mostly summarized by Symons et al. (1998) and come from (1) throughout the southern Canadian Rocky Mountains (Enkin et al. 1997, 1998; Symons et al. 1997, 1998), (2) within the present-day WCSB itself (Cioppa et al. 1998; Lewchuk et al. 1998; Lonnee et al. 1998), and (3) the eastern margin of the basin at Pine Point (Symons et al. 1993). From this direct dating of dolomites and dolomitic limestones, it is becoming very clear that the WCSB underwent a massive fluid flow event in Laramide time that was evidently related to orogenesis (Symons et al. 1999), as Garven (1985) first suggested through paleohydrogeologic modeling.

Petrography and geochemistry

Earlier work by numerous researchers on the diagenesis and dolomitization of the Keg River Formation from the Rainbow area has documented a variety of diagenetic minerals that were formed under a wide spectrum of diagenetic conditions. The matrix dolomite has been interpreted to be formed from marine fluids during a shallow burial stage by an evaporative reflux mechanism (e.g., Langton and Chin 1968; Dunsmore 1971; Schmidt et al. 1985), whereas Qing and Mountjoy (1989) suggested that most of the matrix dolomite was formed from basinal fluids during intermediate burial. The formation of later dolomite and saddle dolomite has been attributed by Qing and Mountjoy (1989) to be the result of deep burial compaction from basinal fluids that were expelled during the Cretaceous prior to oil generation. They based this interpretation on the negative oxygen isotopes and on the high Mn and Fe concentrations found in the saddle dolomite. Similar saddle dolomite is associated with sphalerite and galena in the Presqu'ile barrier at Pine Point and has been interpreted by Qing and Mountjoy (1992) to have resulted from hot, radiogenic, basinal fluids related to tectonic thrusting and compression, sedimentary loading, and tectonic uplift during the Laramide Orogeny. In contrast, Nesbitt and Muehlenbachs (1994) proposed that the Pine Point deposits were formed during the Late Devonian to Early Mississippian from warm basinal brines. Aulstead and Spencer (1985) suggested, based on fluid inclusion evidence, that the dolomite, anhydrite, and fluorite cements in the Keg River Formation of northwestern Alberta were formed from migrating residual evaporitic brines before the end of the Paleozoic.

In the present study, the petrographic and isotopic evidence point to major recrystallization of earlier formed dolomite and the formation of late dolomite from fluids of

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comparable composition. Petrographic and geochemical evidence for this recrystallization includes (1) the increase in crystal size with an increase in nonplanar crystal boundaries (e.g., Sibley and Gregg 1987; Montanez and Read 1992; Al-Aasm and Packard 2000), (2) the irregular contacts between dolomite cores and rims, and (3) the negative shifts in δ^{18} O values. The overlap in isotopic composition of the various types of dolomite (Fig. 7) and the negative δ^{18} O values of –10.7 to –16.5 ‰ support this interpretation. Our data differ from those given by Qing and Mountjoy (1989). In their data, the earlier dolomite has more enriched δ^{18}O values than the microsampled dolomite analyzed in this study. We suggest that early-formed dolomite has been recrystallized by later, probably hot, fluids that precipitated the saddle dolomite. The timing of this fluid flow event would have been during the later phases of the Laramide deformation events, as shown by the paleomagnetic dating of these dolomites.

If we accept the burial history curve for Rainbow buildups from the same area given by Mountjoy et al. (1997), the maximum burial for the Keg River Formation was about 3500 m and occurred in the early Tertiary. If we accept a geothermal gradient of 30°C/km and a surface temperature of 10–20°C (cf. Aulstead and Spencer 1985), the burial temperature would have been in the range of 95 to 105°C. These estimated temperatures are very close to the measured fluid inclusions temperatures in both recrystallized matrix dolomite and saddle dolomite (Table 1), suggesting that recrystallization of fine matrix dolomite occurred by the fluids that precipitated saddle dolomite. However, based on the estimated salinity values obtained form both cores and rims in saddle dolomite, fluids evidently changed in composition as crystallization of saddle dolomite progressed. This is true if we accept that the rims in saddle dolomite formed later than the cores and these later fluids have temperatures similar to those that were involved in the recrystallization of matrix dolomite. The anhydrite was probably precipitated from the same fluids that formed the cores of saddle dolomites. Utilizing the average homogenization temperature measured from dolomite types (ca. 100°C; Table 1), taking an average δ^{18}O value for all dolomite at –12.1‰, and applying the temperature-water isotopic fractionation equation by Land (1983), an estimated $\delta^{18}O_{fluid}$ of ca. –2‰ SMOW was obtained (cf. Al-Aasm and Lu 1994). This value is comparable to the estimated value for Devonian seawater (Knauth and Roberts 1991). This interpretation suggests that deep basinal fluids of marine parentage were responsible for the recrystallization of early dolomite and subsequent formation of later saddle dolomite. This is also supported by the slight enrichment of Sr isotopes in both matrix dolomite and saddle dolomite (Fig. 8).

Conclusions

Based on petrographic evidence, four types of dolomite are recognized in the specimens: (1) a subhedral to anhedral 60–100 μm matrix dolomite that has replaced virtually all primary constituents, (2) a later subhedral 200–400 μm matrix dolomite, (3) a coarse 100–200 μm vug-rimming dolomite, and (4) a late 1000 μm hydrothermal saddle dolomite. All have similar δ^{18} O values between –10.7 and –16.5 ‰ PDB and δ^{13} C values between +0.7 and +3.2 ‰ PBD. The δ^{18} values deviate from postulated carbonate and seawater values for Middle Devonian, which indicates recrystallization of earlier formed dolomite and the formation of late dolomite at elevated temperatures.

AF and thermal step demagnetization isolate a ChRM in specimens from three bore cores in Middle Devonian dolomite strata of the "A" pool of the Rainbow Field. The specimens have negligible VRM and could only be oriented with respect to their core axis; however, the cores were drilled in widely divergent directions. Using the mean ChRM direction to core axis direction angle, a small circle can be drawn for each core, and the three small circles from cores will intersect at the true best-fit direction for ChRM. It is at D = 168°, I = –73.5 ° (α_{95} = 5.8°, k = 32.2) that locates the pole position at 164°E, 83°N (A_{95} = 10°). This pole defines a Tertiary age, probably a Middle Eocene to Middle Miocene age, for magnetization of the dolomites.

We interpret the petrographic, geochemical, and paleomegnetic data to show that deep basinal brines of pre-Laramide marine origin were heated by burial during the Laramide Orogeny. These fluids resulted in the recrystallization of preexisting dolomites and precipitation of saddle dolomite cement during the terminal stages of the Laramide Orogeny. This constrains petroleum migration and accumulation in the dolomite trap to be no older than Eocene at most.

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